Federico Bruzzone ✧ Curriculum Vitae

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Presently a PhD Candidate in Computer Science. I have a strong passion for (AI) **Compilers** and **Programming Languages**. I'm an active contributor to the **MLIR**, **LLVM**, and **IREE** ecosystems, with a strong interest in Rust. For more information, just visit my personal **webpage**.

Scientific Publications

Sep 2026 **Prioritizing Configuration Relevance via Compiler-Based Refined Feature Ranking**
Federico Bruzzone, Walter Cazzola, and Luca Favini
Journal of Systems and Software, Elsevier, 113096
Q1 Scimago | bib | pdf | SpringerLink | arXiv

Sep 2026 **The Minimal Essence of Higher-Order Functions in Maude**
Federico Bruzzone, Lorenzo Capra, Walter Cazzola, and Carlos Olarte
Proceedings of the 28th Int. Symposium on Symbolic and Numeric Algorithms for Scientific Computing (SYNASC'26), IEEE Computer Society
C on CORE | bib | pdf

2026 **Symmetry Preservation in Modular Rewritable Multiformalism Models**
Lorenzo Capra and **Federico Bruzzone**
Joint Workshop Proceedings of PNSE'26, PNAS'26, ATAED'26 and PHOCON'26, CEUR Workshop Proceedings, vol. 4236
B on CORE | bib | pdf | CEUR-WS

2026 **Sink or SWIM: Tackling Real-Time ASR at Scale**
Federico Bruzzone, Walter Cazzola, Matteo Brancaleoni, and Dario Pellegrino
IEEE Transactions on Audio, Speech and Language Processing, pp. 1–11
Q1 Scimago | bib | pdf | IEEE | arXiv

Sep 2025 **Code Less to Code More: Streamlining Language Server Protocol and Type System Development for Language Families**
Federico Bruzzone, Walter Cazzola, and Luca Favalli
Journal of Systems and Software, Elsevier, 112554
Q1 Scimago | bib | pdf | SpringerLink | arXiv

Preprints

Jun 2026 **A Multi-Dimensional, Per-Pass Empirical Study of the LLVM Optimization Pipeline**
Federico Bruzzone and Walter Cazzola
arXiv:2606.31238 [cs.SE] arXiv

May 2026 **Generalized Software Product Line Extraction**
Federico Bruzzone, Walter Cazzola, and Luca Favalli
arXiv:2605.28989 [cs.SE] arXiv

Feb 2026 **Meta-Monomorphizing Specializations**
Federico Bruzzone and Walter Cazzola
arXiv:2602.12973 [cs.PL] arXiv

Feb 2026 **From Separate Compilation to Sound Language Composition**
Federico Bruzzone, Walter Cazzola, and Luca Favalli
arXiv:2602.03777 [cs.PL] arXiv

Education

2024–Present PhD Candidate in Computer Science at the ADAPT Lab, University of Milan

2022–2024 MSc in Computer Science at University of Milan (*110/110 cum laude*, W. Cazzola, 15/07/2024)
Thesis: Toward a Modular Approach for Type Systems and LSP Generation

2019–2022 BSc in Musical Computer Science at University of Milan (13/10/2022)

2011–2019 Piano and Music Composition at I.S.S.M. Novara Conservatory

2014–2019 Diploma in Computer Science and Telecommunications at E. Alessandrini

Open Source Contributions

2024–Present **MLIR/LLVM** **48** PRs (**38** merged) · **150** reviews · **9** issues

RFC **Unify masking in the Vector dialect** · *LLVM Discourse*

Poster **A Multi-Dimensional, Per-Pass Empirical Study of the LLVM Optimization Pipeline** · *2026 US LLVM Developers' Meeting*

RFC **Transformation support for `linalg.pack`/`linalg.unpack` on memrefs** · *LLVM Discourse*

RFC **Improving PR discoverability for aspiring reviewers** · *LLVM Discourse*

2026–Present **IREE** **6** PRs (**5** merged) · **7** reviews · **8** issues

2026–Present **Source Matching and Rewriting** **15** PRs · **1** review

2026 **LLVM Lighthouse** **2** PRs (**1** merged)

2024–2025 **Rust** **4** PRs (**2** merged) · **2** reviews · **1** issue

2025 **rustworkx** **1** PR · **1** issue

From 2026 The Eter Programming Language, a new programming language leveraging the MLIR/IREE ecosystem with tensors support.

From 2026 Maintainer of the LLVM Passview, a framework to run an incremental, reproducible study of LLVM optimization passes.

From 2026 Maintainer of the LLVM Pass Template, a C++ template project to quickly create new LLVM passes.

From 2025 Maintainer of the Papers on Compiler Optimizations: Analysis and Transformations, a curated list of scientific publications.

From 2025 Maintainer of scribe, a minimalist, opinionated LaTeX document class for technical writing and presentations.

From 2025 Maintainer of the Tide Compiler, a backend-agnostic IR and compiler framework written in **Rust**.

From 2024 Maintainer of the cross-platform tgt project and tdlib-rs TDLib bindings written in **Rust**.

From 2024 CHIP-8 STM32-Port: CHIP-8 and S-CHIP emulator ported to STM32 Cortex M4 microcontroller.

From 2023 CHIP-8 Core: Core CHIP-8 and S-CHIP emulator library, used as the foundation for platform ports.

Research Activities

Oct 2026 **Reviewer** for the Journal of Computer Languages (COLA), *Elsevier*, C on CORE

May 2026 **Committee Member** at the Int. Conf. on Software Language Engineering (SLE 2026), *ACM*, B on CORE

Dec 2025 **Reviewer** for the Eur. Conf. on Object Oriented Programming (ECOOP 2026), *ACM*, A on CORE

Nov 2025 **Reviewer** for the Journal of Computer Languages (COLA), *Elsevier*, C on CORE

Jun 2025 **Reviewer** for the Journal of Software and Systems Modeling (SoSyM), *Springer*, Q1 on Scimago

2–6 Jun 2025 **Participant** at <Programming>2025 conference

Apr–Jun 2025 **Committee Member** at the Int. Conf. on Software Language Engineering (SLE 2025), *ACM*, B on CORE

Mar 2025 **Reviewer** for the Journal of Systems and Software (JSS), *Elsevier*, Q1 on Scimago

Sep 2024 **Student Volunteer** at the Int. Conf. on Functional Programming (ICFP 2024)

Dissemination Activities

Oct 2026 A Multi-Dimensional, Per-Pass Empirical Study of the LLVM Optimization Pipeline — Poster at the 2026 US LLVM Developers' Meeting, Santa Clara.

Jun 2026 A Multi-Dimensional, Per-Pass Empirical Study of the LLVM Optimization Pipeline — Ph.D. Student's Activity Report, University of Milan.

Mar 2026 MLIR: Scaling Compiler Infrastructure for Domain Specific Computation — Presentation of the paper by Lattner et al., 2021.

Feb 2026 Matheuristic Variants of DSATUR for the Vertex Coloring Problem — Presentation of the paper by Dupin, 2024.

Dec 2025 Your Optimizing Compiler is Not Optimizing Enough. To Hell With Multiple Recursions! — First MuseMI meeting of the 2025/2026 season, Milan.

Jul 2024 P4 Compiler in SDN — Presentation on the P4 compiler in Software Defined Networking.

Jul 2024 PhD Project — *Universal Language Server Protocol and Debugger Adapter Protocol for Modular Language Workbenches*, University of Milan.

Jul 2024 Master's Thesis — *Toward a Modular Approach for Type Systems and LSP Generation*, University of Milan.

Teaching Activities

Thesis Supervision

14/07/2026 M. Visconti, *Towards Neverlang3: A Generative Rust Approach for Resilient LL Parser Generators*, MSc, **102**

23/02/2026 D. Cerato, *Employing Metaprogramming: A Macro-Driven Strategy to Overcome Compiler-Imposed Limitations on Specialization*, BSc, **103**

10/04/2025 D. Pellegrino, *Scalable Multi-client Real-time Whisper*, BSc, **96**

09/04/2025 A. Longoni, *GUIDE: Graphical User Interface Development Environment*, MSc, **110L**

09/04/2025 L. Albani, *New Generalized Protocol For Software Product Line Extraction And Configuration*, MSc, **110L**

09/04/2025 G. Esposito, *Fr3D: A Framework for DAP-compatible DSL-oriented Debugging*, MSc, **110L**

24/02/2025 L. Favini, *RustyEx: Instrumenting* `rustc` *to Extract Feature Dependency Graphs*, BSc, **102**

Graduate Courses

2026–2027 Mathematical Logic (Art. 45), **BSc** in CS, University of Milan, *S. Aguzzoli*

2025–2026 Mathematical Logic (Art. 45), **BSc** in CS, University of Milan, *S. Aguzzoli*

2024–2025 Programming 1 (Art. 45), **BSc** in CS, University of Milan, *L. Capra* (coordinator W. Cazzola)

2024–2025 Mathematical Logic (Art. 45), **BSc** in CS, University of Milan, *S. Aguzzoli*

2023–2024 Mathematical Logic, **BSc** in CS, University of Milan, *S. Aguzzoli*

2023–2024 Computer Science, **BSc** in Comm. and Society, University of Milan, *A. Momigliano*

2023–2024 Programming 1, **BSc** in CS, University of Milan, *A. Trentini* (coordinator P. Boldi)

2023–2024 Programming in Python, **MSc** in Chemistry, University of Milan, *M. Monga*

Additional Activities

2023–Present **Private Tutoring** in Computer Science, Mathematics and Physics

Nov 2024 **Collaborator** at the Bebras Challenge, ALaDDIn Lab

Jan–Jun 2024 **Organizer** of the BSc Computer Science Laboratories, ALaDDIn Lab

Jan–Jun 2024 **Collaborator** of the Workshops for schools, ALaDDIn Lab

Nov 2023 **Collaborator** at the Bebras Challenge, ALaDDIn Lab

Musical Activities

2021–Present **Sound Engineer and Music Producer**, in collaboration with Believe and as a SIAE member.

2006–Present **Pianist and Music Composer**, composing original pieces and arrangements for piano and various ensembles

2019–Present **Piano and Music Teacher**, teaching piano and music theory and composition to students of all ages and levels

Grants and Fellowships

2024 56th Top Github Public Contributor in Italy out of 958

2023–2024 Scholarship for the MSc in Computer Science, awarded by the University of Milan

2020–2024 Scholarship for the BSc in Musical Computer Science, awarded by the University of Milan

Tongues

Italian Mother tongue

English Level CEFR B2 (SLAM at University of Milan)

Spanish Base (A1–A2)

Milan, November 1, 2026